“Why the face?”: Exploring Robot Error Detection Using Instrumented Observer Reactions

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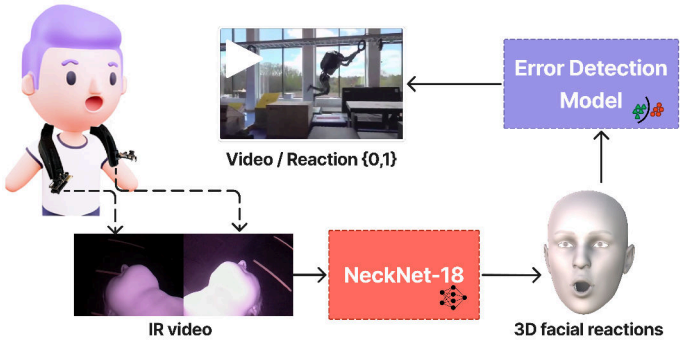


Fig. 1. User study scheme. Participants wearing NeckFace [8] watch videos where a scenario of human or robot error is shown, eliciting a reaction. The IR camera image is converted into 3D facial points and head rotation data through a customized NeckNet model. This data is then used to train error detection models which map human reactions to the scenario displayed.

How do humans recognize and rectify social missteps? We achieve social competence by looking around at our peers, decoding subtle cues from others around us — a raised eyebrow, a laugh — to evaluate the environment and our actions. Robots, however, struggle to perceive and make use of these nuanced reactions. By employing a novel neck-mounted device that records facial expressions from the chin region, we explore the potential of previously untapped data to capture and interpret human responses to robot error. First, we develop NeckNet-18, a 3D facial reconstruction model to map the reactions captured through the chin camera onto facial points and head motion. We then use these facial responses to develop a robot error detection model which outperforms standard methodologies such as using OpenFace or video data, generalizing well especially for within-participant data. Through this work, we argue for expanding human-in-the-loop robot sensing, fostering more seamless integration of robots into diverse human environments, pushing the boundaries of social cue detection and opening new avenues for adaptable robotics.

Additional Key Words and Phrases: wearable; robot error; social signals; HRI; computer vision; human-AI collaboration

1 Introduction

Humans navigate social environments by perceiving and interpreting cues from those around them. *Social competence* stems partly from this ability to read non-verbal signals such as facial expressions, body language, and subtle gestures. These environmental and peer-based inputs form a feedback loop that informs social behavior and facilitates integration into diverse social contexts [18].

The concept of *human-in-the-loop sensing* in robotics leverages this innate human capability to enhance the social competence and efficiency of robotic systems. This approach posits that human reactions can serve as a rich, real-time source of information for robots, guiding their behavior and functionality. For example, methods for Reinforcement Learning from Human Feedback (RLHF) have leveraged human social cues and emotions as feedback input [26]. Recent works [34, 39, 42] have started exploring the potential of human reactions in the context of human-robot interaction (HRI) for robot error detection. These studies make use of human social cues, captured through various means, as indicators of robot performance and social appropriateness.

Traditional implementations of this concept often rely on stationary cameras or extensive sensor arrays to capture human reactions. While groundbreaking, these methods can be resource-intensive, computationally demanding, and may struggle to adapt to dynamic, real-world environments where both humans and robots are in constant motion. To address these limitations, we employ NeckFace [8], a novel, neck-mounted device that records facial expressions from the chin region and maps them onto 3D facial points and head rotation. This lightweight, bi-camera system offers an alternative perspective on human reactions, potentially capturing nuanced expressions that may be overlooked by traditional methods. Its wearable nature allows for data collection in dynamic, real-world scenarios, providing a more efficient and adaptable solution for understanding human responses to robot behavior. A neck-mounted, wearable sensor is intended as a complement to, rather than a replacement for, a robot's own onboard cameras: it can capture a user's facial reactions in situations where the robot cannot see the user's face (e.g., when the user is behind or beside the robot, when the robot lacks a camera oriented toward the user, or in multi-robot and non-instrumented environments where relying on infrastructure cameras is impractical). Because the sensor moves with the wearer, it also preserves reaction tracking during locomotion or transient occlusion, conditions under which fixed or robot-mounted cameras can lose line of sight to the face.

We note that the present study evaluates NeckFace in a controlled, stationary, video-elicitation paradigm, rather than during an in-person, mobile human-robot interaction. We adopted this paradigm as a first, controlled step that allows us to benchmark NeckFace's facial reaction data against established alternatives (e.g., RGB video and OpenFace features) under matched stimuli and conditions, before testing the device in a less-controlled, in-the-wild deployment. We revisit this scoping decision, and its implications for the motivation of the work, in Section 6.4.

To test and benchmark the efficiency of NeckFace against other conventional approaches, we conducted a comparative user study. Participants' reactions to video stimuli displaying human and robot error were recorded simultaneously using NeckFace and conventional RGB cameras. Following this stage, we 1) developed and tested a model to map NeckFace's IR camera data onto 3D facial expressions and head motion data, NeckNet-18; 2) used this reaction data to develop error detection models that perform well for unseen participants but also generalize to a single participant with minimal data for training; and 3) compared these models with conventionally used methods, such as OpenFace features [3, 39, 42] or RGB camera data [4, 33], finding that it outperforms these methods.

This research not only contributes to the development of more dynamic and context-aware robotic systems but also advances our understanding of how human social competence can be effectively leveraged in HRI. By exploring

alternative data sources and expanding on human-in-the-loop sensing methodologies, we aim to pave the way for more socially competent and efficient robotic systems that can seamlessly integrate into human environments.

2 Related Work

Social cues for error detection: Beyond the HRI literature, a longstanding line of work in social psychology has studied how people react, facially and behaviorally, to social transgressions and missteps – whether their own or someone else's. Embarrassment, in particular, has been characterized as having a distinct, reliably recognizable nonverbal display (gaze aversion, head movement down and to the side, a controlled or suppressed smile, and face touching) that functions as a form of appeasement following a social norm violation, helping to restore social relations between the person who erred and observers [20]. This display has also been shown to be empirically distinguishable from the facial signatures of related self-conscious emotions, such as shame and guilt [21]. This literature offers a theoretical lens for interpreting facial reactions to robot error: the same appeasement and self-conscious-emotion displays studied in human-human social transgressions may plausibly extend to how observers react when a robot, rather than a person, makes a mistake.

Human-in-the-loop robotic systems [38] leverage human input to enhance robots' efficiency, yielding positive outcomes in HRI [11]. Following this concept, harnessing human social cues can equip robots with better social and operational competence. For example, a substantial body of literature has begun exploring social cues in response to robot error, revealing that human reactions are multimodal and diverse [7, 34], including body and head motion [12, 22, 43], gaze [6], and facial expressions [1, 16, 22, 41]. The detection of robot error from social cues has been investigated through various machine learning approaches. Stiber et al. [42] proposed a framework for "error-aware " HRI, employing a deep neural network, building on previous work [40]. Other studies [34, 39] utilize recurrent neural networks (RNNs) to preserve time-dependencies. Random Forest classifiers have also demonstrated efficiency as linear methods for error detection [29]. Recently, combinations of Minirocket [10] classifiers have achieved high performance in error and confusion detection [44, 45]. To bypass the need for feature extraction, some researchers have employed convolutional neural networks [4, 33] to classify human reactions directly from video data. Recent work by Ravishankar et al. [36] explores unsupervised learning of robot error by detecting abnormalities in human-robot interactions from human-human interaction data. These works explore visual, audio and semantic data modalities. Given the novelty of the field, however, there remains untapped potential for exploring novel data modalities and processing methodologies for social cue classification.

Beyond standard methods for data collection: Capturing facial expressions as humans watch robots can allow the robot to infer human perceptions of robot performance and social appropriateness. High-accuracy facial expression tracking traditionally involves active motion capture instrumentation on the face, either using visual markers [37] or other sensors such as electromyography (EMG) [13] or capacitive sensors [35]. These methods usually require extensive setups and calibration and can cause physical and social discomfort. Relieving users from heavy on-face instrumentation, frontal-camera-based vision solutions have been getting recent attention [2, 30]. These solutions usually involve placing a camera in front of the user's face to directly record and analyze their facial expressions. While they provide a less intrusive experience, they are not mobile, as users have to appear in front of the camera within reasonable angle and distance. To address these challenges, recent wearable solutions look into less intrusive sensing solutions while trying to maintain tracking performance [8, 24, 25, 46].



Fig. 2. Study setup. In the *Calibration round*, the participant, wearing NeckFace [8], copies the movements seen on a video on an Iphone 11. The *Stimulus round* consists of a series of 30 videos played on a screen, while a webcam and NeckFace record facial reactions.

Among these, NeckFace [8] is a wearable facial expression tracking system that can estimate 3-dimensional facial expressions continuously. It performs comparably to traditional video-based systems, without the need for a frontal camera. Unlike other mounted devices, NeckFace is robust to users' head rotations, allowing for greater flexibility in use. It features a 3D-printed neckband with two infrared (IR) cameras positioned on either ends pointing upwards toward the wearer's neck, chin and cheeks (Figure 1). Two 850 nm IR LEDs are placed beside each camera to provide independent lighting to minimize the impact of unstable environmental lighting. NeckFace uses a convolutional neural network (ResNet-34 [15]) to convert the captured images into 3D facial expressions represented by 52 Blendshapes. NeckFace has been demonstrated to have reliable tracking performance across different scenarios such as sitting, walking and remounting. In light of this, we chose NeckFace as our facial reaction reconstruction device.

To the best of our knowledge, this is the first user study pursuing the use of a neck-mounted device to capture human social cues in HRI, investigating and benchmarking its potential for error detection.

3 Data Collection

In this section, we detail the NeckFace system implementation, user study design and methodology for data collection. To collect human reactions to observed errors, we followed a procedure adapted from Bremers et al. [4] for in-lab data collection, where participants observed a set of videos where error occurs. The elicitation materials and code used are shared in the project repository [1].

3.1 Sensors and System Implementation

In this work, both IR cameras from NeckFace are connected to a single Raspberry Pi 4B via an Arducam multi-camera adapter, different from the original NeckFace system which uses two Raspberry Pis to record the images [8]. This allows for a lighter system that is easier to wear and integrate with other components. The two cameras capture images of the wearer's neck, chin and cheeks from both sides, which are subsequently used to estimate the full facial expressions in 3D.

[1] http://irl.tech.cornell.edu/badrobots-feat-neckface/

In addition to the NeckFace device and calibration setup, a participant-facing webcam was used to record participants' reactions.

3.2 Study Protocol

The study set up is shown in Figure 2. Participants were recruited through flyers and word-of-mouth. After giving informed consent and providing demographic information, participants were taken to the experiment room, where the researcher mounted NeckFace on them, adjusting the positioning of the cameras as necessary. Following this, participants completed a *calibration round*, where they were placed in front of an iPhone device and asked to copy the movements seen on a video, such as moving their head up and down or raising their eyebrows (approximately 5 minutes), complying with the initial NeckFace work [8]. After this, the researcher would leave the room and data collection would begin, as participants completed a *stimulus round* where they watch a series of 30 videos presented in random order on a screen. Participants watched each stimulus video while a high-resolution webcam and NeckFace recorded their facial responses. Participants were not able to see their own image while the stimulus videos played. Compensation was provided at rate of USD 15/hour. After this, participants were taken back to a smaller room to fill out a final written questionnaire. The full procedure took around 30 minutes to complete. A diagram of study stages and data collected can be seen on Figure 3. This data was collected under Cornell University IRB exempt protocol #1609006604.

3.2.1 Calibration Data Collection. The NeckFace system needs to be trained in order to provide accurate facial expression estimations. To supervise the training process, a dataset of (IR camera images, 3D facial expression) pairs needs to be collected. Specifically, the 3D facial expressions (ground truth) were represented by 52 Blendshapes [37] captured with the TrueDepth camera on an iPhone 11 Pro. We also predicted three head rotation variables. We obtained the training data through the calibration round, where participants were asked to follow movements and facial expressions seen on a video (Figure 2).

3.2.2 Stimulus Data Collection. We selected a set of 30 stimulus videos for collecting human reactions. Following the methodology in Bremers et al. [5], we define errors as "actions not as intended". For example, a robot falling while playing soccer, or a person crashing a lawnmower onto an object. We included videos of human failures (10), robot failures (10) and control videos (10) (see Figure 2 for example). The videos' average length is 13.69 ± 7.77 s. The full list can be consulted in the project repository [1].

3.3 Participants

A total of 30 participants completed the study. Due to technical issues or poor quality of data, 5 participants were excluded (P17 and P27 were removed because they were wearing large earrings which would shake and cause disturbances in the image quality; P27, 29 and 30 for inadequate NeckFace camera adjustment; and P13 due to erratic head motion during calibration). The remaining 25 participants took on average 33m22s±4m19s to complete the survey. Ages range from 21-78 (37.73 ± 17.37). 17 participants identify as female, 12 as male, and 1 as non-binary. The study included participants from 11 nationalities. Racial/ethnical distribution includes 17 Caucasian/White or Asian/White, 6 Asian/Asian American participants, 4 Hispanic/Latino and 1 African/African American/Black, 2 participants who self-described as Indian and Indian American.

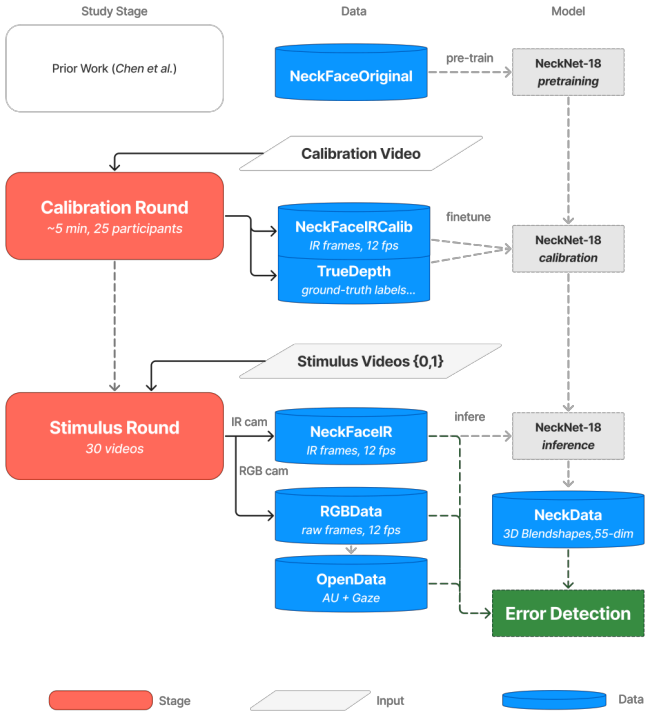


Fig. 3. Study protocol and data collected. In the calibration round, NeckFace IR camera data (NeckFaceIRCalib) is collected alongside TrueDepth data from the iPhone (including head rotation angles), which serves as ground truth to train NeckNet-18. NeckNet-18 is first pre-trained on NeckFaceOriginal, a dataset from prior work [8], then fine-tuned using calibration data from this study. In the stimulus round, reactions to neutral (0) and error (1) videos are recorded simultaneously through the NeckFace IR cameras (NeckFaceIR) and a participant-facing RGB camera (RGBData). NeckFaceIR is passed through NeckNet-18 to produce NeckData, a dataset of 3D facial reactions. RGBData is processed in parallel through OpenFace [3] to produce OpenData, a dataset of action unit and gaze features. NeckData and OpenData serve as the primary inputs to the error detection models evaluated in Section 5; NeckFaceIR and RGBData are additionally tested as direct frame-based inputs.

3.4 Overview of Datasets

Added this section.

Several distinct datasets are referenced throughout this paper, each derived from the calibration or stimulus rounds described above, and each serving a different purpose in model training and evaluation. Table 1 summarizes all of them in one place for reference; each is described in more detail in the section indicated.

4 Proposed System

4.1 3D Facial Mapping Model

The 3D facial mapping model takes IR video frames captured by the NeckFace device and converts them into numerical representations of 3D facial expressions. The model was developed following the original design of NeckNet [8].

Problem Formulation and State space: Tracking facial expressions with the wearable neckband was formulated as a regression problem: at any given time, the raw data from the wearable device is captured as two images $i_l, i_r \in \mathbb{R}_{640\times480}$,

Table 1. Overview of all datasets referenced in this paper.

Dataset	Source / collection method	Rate	Used for
NeckFaceOriginal	IR camera frames from the original NeckFace study [8] (13 participants)	30 fps	Pretraining NeckNet-18 (subsection 4.1)
NeckFaceIRCalib	NeckFace IR camera frames, calibration round (this study)	12 fps	Fine-tuning/evaluating NeckNet-18 (subsection 4.1)
NeckFaceIR	NeckFace IR camera frames, stimulus round (this study)	12 fps	Input to NeckNet-18 to produce NeckData; also tested directly as a frame-based error detection input (subsection 5.2)
NeckData	55-dim. Blendshape + head-rotation vectors, reconstructed from NeckFaceIR via NeckNet-18	12 fps	Error detection model training/evaluation (subsection 5.2)
OpenData	OpenFace [3] AU/gaze/pose features extracted from the RGB stimulus-round video	12 fps (downsampled from 30)	Error detection model training/evaluation (subsection 5.2)
RGBData	Raw RGB stimulus-round video frames	12 fps (downsampled from 30)	Frame-based error detection model training/evaluation (subsection 5.2)

from the left and right side cameras, respectively. The model output is a feature vector $e \in \mathbb{R}_{55\times1}$, representing the 3D facial expression and head orientation.

Action Space : The model output $e \in \mathbb{R}_{55\times1}$ represents the 3D facial expression. This includes 52 Blendshape parameters extracted from TrueDepth camera by ARKit, representing the frontal face, and 3 angles representing the orientation (yaw, pitch, roll) of the head. The 52 Blendshapes each represent a specific facial movement (e.g., jawOpen, mouthClose, etc.), with numerical values linearly describing the extent of that movement on a scale of 0 to 1000. The ground truth of these Blendshapes was captured with the TrueDepth Camera of an iPhone 11 at 30 fps and later resampled to 12 fps to match the sampling rate of the IR cameras. A full list of the 55 features is provided as supplementary material.

Dataset : Two datasets were used to train and evaluate the facial reconstruction model. In addition to the data collected in this study, we used the original NeckFace dataset [8] for training.

- **NeckFaceOriginal**: video frames collected for Chen et al. [8]. It includes images recorded from two IR cameras at 30 fps from 13 participants. This dataset was used to pre-train the model.
- **NeckFaceIRCalib**: video frames from NeckFace's IR cameras collected during the calibration round. Frames were recorded at 12 fps. This dataset was used to fine-tune the pre-trained model. See Figure 3 for a schematic of how this data was used.

Data processing and augmentation were applied both during pre-training and fine-tuning. Augmentation was used to improve the model's robustness and generalization given the relatively limited amount of calibration data available per participant, and to make the model more tolerant of small, expected variations in camera position, head pose, and lighting that occur naturally as the device is worn and re-worn. Specifically, raw images from the two cameras were first converted to grayscale and resampled to a size of 320 by 240 pixels, then tilted horizontally to generate an image of size 640x240 pixels. Data augmentation methods include: 1) 50% chance of random scaling with a factor of 0.9 to 1.1, 2) 50% chance of random rotation of -30 to +30 degrees (while pre-training) or -8 to +8 degrees (while fine-tuning), 3) 50% chance of random translation of -6 to +6 pixels in both directions. These operations were applied to both sides of the images independently but with the same parameters.

Evaluation Metrics : The model was evaluated with the Mean Absolute Error (MAE) between the predicted Blendshapes and ground truth (TrueDepth). The error of the facial expressions and head orientation were calculated separately.

Specifically, $\text{MAE}_f = \sum_{i=1}^{52} \frac{1}{52}|e_i - \hat{e_i}|$, $\text{MAE}_o = \frac{1}{3}\sum_{i=53}^{55}|e_i - \hat{e_i}|$ where MAE_f is the error of facial expressions tracking and MAE_o is the error of head orientation tracking. e is the ground truth of facial expression and head orientation, and $\hat{e}$ is the model prediction.

Model Design : We designed NeckNet-18, a convolutional neural network (CNN) with a ResNet-18 backbone and a fully-connected decoder (Figure 4). We based this structure on the previous work [8], with adaptations to better fit our data and task. First, we used a lighter-weight ResNet-18 instead of the ResNet-34 model used in prior work [8], based on two considerations: 1) the dataset collected in this user study is smaller than that in Chen et al. [8] due to smaller calibration time and a lower frame rate. Larger models would be more prone to overfitting. 2) ResNet-18 has roughly half the number of trainable parameters when compared with ResNet-34, which results in faster training and inference speed. Previous work on similar tasks also demonstrated that ResNet-18 has comparable performance to ResNet-34 with a smaller dataset [25].

Model Training : NeckNet-18 is designed for within-participant generalization, using a small Calibration round to fine-tune the model for new participants. Thus, the model's generalizability relates to unseen *temporal* data rather than unseen participants.

We pre-trained the model on the *NeckFaceOriginal* dataset. To fine-tune and evaluate the performance of the 3D facial mapping model, we used data collected from the calibration round, *NeckFaceIRCalib*. Specifically, we first divided this dataset into 5 folds by dividing data collected from each participant's calibration round into 5 parts *in temporal order*, and combining each of the 5 parts from different participants together. This means that each fold contains data from all participants, but each fold's data does not overlap temporally. We then chose 4 of the 5 folds to use as training data to fine-tune the model, and the other fold as testing data. This process was repeated 5 times, constituting a 5-fold cross-validation where all 5 folds had appeared in the testing data once. This step allowed for hyperparameter tuning and also allowed us to understand the within-participant generalizability of the model. Once the best combination of hyperparameters was found, all 5 folds were combined and the model was trained again to maximize the benefit of all available training data.

During pre-training, the model was trained for 30 epochs with an initial learning rate of 0.0002. Given the differences in camera angle and image quality between the original NeckFace implementation and our re-implementation, the fine-tuning step which followed was given more weight to allow the model to better learn the distribution of the new dataset. As a result, during fine-tuning, the model was also tuned for 30 epochs but with an initial learning rate of 0.0001.

After training, the model was used to map (i.e., predict) 3D facial expression and head-rotation values from new, previously unseen NeckFace IR frames – specifically, the frames collected during the *stimulus round* (participant reacted to video scenarios), producing the corresponding 3D facial reaction values for each frame. This generated the dataset of 3D facial reactions (NeckData) used as input for the error detection model (Figure 3).

5 Evaluation

5.1 3D Facial Mapping Model: Evaluation

Following the procedures described in Section 4.1, we ran 5-fold cross-validation (non-overlapping temporal data) to test NeckNet-18 performance for 3D facial mapping. For Blendshape parameter data (facial motion), performance

was $\text{MAE}_f = 34.0 \pm 6.9$ ($M \pm SD$), and $\text{MAE}_o = 7.0 \pm 1.3$ for head rotation angles. For reference, the original NeckNet performance [8] was $\text{MAE}_f = 25.6 \pm 5.1$, $\text{MAE}_o = 3.1$.

5.2 Error Detection Model

We used the output of NeckNet-18 (55 Blendshape and head motion parameters) while participants were watching stimulus videos and trained models that *detect when humans are reacting to an error*. We benchmarked this performance against other conventional data inputs, and tested the models' ability to generalize to new participants or within the same participant data. Figure 3 illustrates how study stages and datasets are integrated into our processing methodology.

Problem Formulation: Detecting error based on human reactions was formulated as a sequential decision-making problem: at each time step t, the environment is captured as a state variable $s_t \in S$ and the model output is an outcome label $f_t \in F$, where F is a discrete variable that describes the stimulus: 0 if *neutral*, 1 if *error*.

Action Space: The reactions captured with NeckFace IR cameras and RGB video were labeled according to the type of stimulus video shown to participants. In videos where an error occurs, only the moments following the labeled error timestamp for that video (the "Error Time" annotation listed per video in the supplementary material) were used as input data (and labeled *1, reaction to error*); this per-video timestamp marks when the error itself occurs within the stimulus video, and is distinct from the duration of the reaction window used for analysis, which is constant in length and applied uniformly from that point forward. Control videos, where there is no error, were used integrally and labeled *0, neutral state*. This reduced data imbalance across classes.

Datasets: We tested 4 data input sources to test the performance of NeckFace against other commonly used methods (see also the overview in Table 1).

- **NeckData:** This dataset was collected at 12 fps, consisting of 55 Blendshape and head motion parameters as described in Section 4.1, reconstructed using NeckNet-18.
- **OpenData:** Reactions to the stimulus videos were also recorded using a RGB camera at 30 fps. We ran OpenFace [3] on this data and obtained 49 Action Units (AUs), gaze and pose features. The dataset was undersampled to 12 fps, to match the NeckData timestamps.
- **NeckFaceIR:** We used video frames from NeckFace's IR cameras to test direct output classification. Cameras collected data at 12 fps and this data was resized to 224x224. Post-processing and data augmentation were performed on each frame, according to ImageNet [23] statistics for data normalization.
- **RGBData:** We used frames from the RGB camera reaction videos, which were processed as in *NeckFaceIR* and downsampled to match the 12 fps sampling and timestamps.

We also tested normalizing the dataset and running a Principal Component (PC) Analysis for feature reduction. We kept only the PCs which explained 95% of the variance in the dataset. This resulted in 15 features for the NeckData dataset, and 33 features for the OpenData dataset. The total number of individual data points per dataset can be seen in Table 2. Because all participants watched the same stimulus videos, data is balanced across participants.

Models tested: To develop an error detection model from human reactions, we tested a diverse range of model architectures based on prior work in the space of robot error detection [33, 39, 45]. These included Recurrent Neural Networks (RNNs) such as Long Short-Term Memory (LSTM) networks [19], which excel at capturing long-term dependencies; Gated Recurrent Units (GRUs), known for their efficiency in training; and Bidirectional LSTMs (BiLSTMs), which process sequences in both directions. We also tested **GRU_FCN**, a hybrid architecture combining a GRU

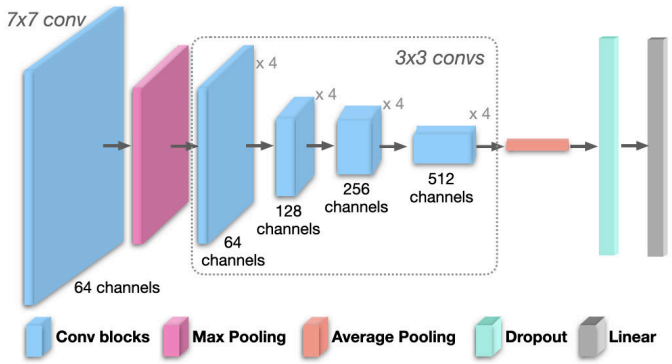


Fig. 4. NeckNet-18 architecture.

branch with a fully-convolutional network (FCN) branch in parallel, whose outputs are concatenated before the final classification layer. We explored Transformer models, leveraging their self-attention mechanism for capturing global dependencies. Additionally, we implemented MiniRocket [10, 44, 45], a non-neural approach known for its speed and accuracy in time series classification. The gated Multi-Layer Perceptron (gMLP) [27] was tested as an alternative to attention-based models, using spatial gating units. Lastly, we included InceptionTime and InceptionTimePlus [17], an ensemble of deep Convolutional Neural Network (CNN) models designed specifically for time series classification. We also tested a similar Deep Neural Network to that suggested by Stiber et al. [42]—3 hidden layers, with 64, 128 and 64 units (multi-layer DNN, ml-DNN). For the video datasets, we used ResNet34 models [14], which are 34-layer CNNs pre-trained on ImageNet [23], and tested different model structures. ResNet34 is the base model for the original version of NeckNet [8].

Evaluation Metrics: The models were evaluated based on the macro averages of the following metrics: accuracy, F1-score, precision, and recall. We also considered margin-of-error metrics [9, 32, 39]—for a sample margin of size k, and for a sample i, the model prediction is considered correct if $y^i_{pred} \in [y^{i-k}_{pred}, y^{i+k}_{pred}]$. Similar to prior works, we consider these metrics to contemplate real-life error detection scenarios, where a small tolerance in timely detection does not invalidate the system.

Model Training: We implemented model training through tsai [31] and pytorch. For all model types and datasets, we performed hyperparameter tuning for a maximum of 500 epochs using a 5-fold cross-validation procedure at the participant level, with a 70-20-10 train-validation-test proportion on each fold. Specifically, the 25 participants were randomly ordered once (fixed seed) and divided into five disjoint groups of five participants. In each of the five iterations, one group (5 participants, 20%) served as the validation set, the next 3 participants in the ordering (≈10%) served as a held-out test set, and the remaining 17 participants (≈70%) were used for training; no participant appeared in more than one of the three sets within an iteration. All metrics reported in Table 3 for cross-participant models are the mean ± SD of test-set performance across these five iterations. For time-series models (all models in this list except the frame-based ResNet34 models below), we implemented an interval length (i.e., number of samples in a training sample window) and stride (i.e., sliding length between sample windows) according to the methodology in Wachowiak et al. [44], since this windowing scheme is specific to sequential/time-series input and does not apply to the frame-by-frame classification setup used for the ResNet34 models. Labels were selected according to the mode (most frequent label) of

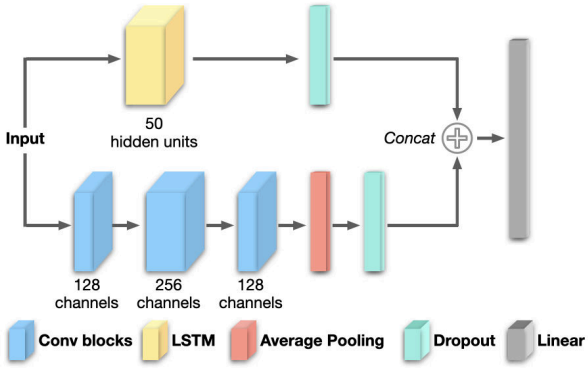


Fig. 5. Best-performing model for NeckData, GRU_FCN.

each sample window. We picked the top-performing hyperparameter configuration of each model type based on average test accuracy across the 5 folds, and, within each configuration, selected the best-performing epoch based on average validation accuracy across the 5 folds. For the ResNet34-based models (used on the NeckFaceIR and RGBData datasets), best-performing models were instead selected based on test accuracy on a single fold, rather than averaged across all 5 folds as for the other model types. This deviation was a practical choice: training ResNet34 on full video frames across all 5 folds was substantially more computationally expensive and time-consuming than training the time-series models, and single-fold selection was used to keep training time tractable. We acknowledge this is a methodological inconsistency relative to the other models and discuss its implications as a limitation in Section 6.4.

We additionally tested cross-training of the top-performing model types. This was motivated by the question of whether transfer learning between the NeckData and OpenData feature spaces could improve performance relative to training on either dataset alone – i.e., whether facial-reaction representations learned from one camera modality transfer usefully to the other [33]. That is, we used the best-performing models on the *NeckData* and *OpenData* datasets, respectively, and fine-tuned the model's last layer to the alternative dataset, following the same process for model selection as described above.

Finally, we tested single-participant generalization, based on minimum data required for good model performance. For this, we used n folds (n = number of participants), and trained the models on each i fold using the first 5%, 10%, 25%, or 45% of participant i's stimulus-round data in temporal order; a further 20% of the training amount was used for validation and all remaining data from that participant was used for testing. We report the two extremes of this range (5% and 45%) in Table 3, since they respectively characterize the low-data regime that motivates a wearable approach and the performance ceiling as per-participant data increases.

Final Dataset for Error Detection: The size of the datasets used for training is presented in Table 2. NeckData is obtained by reconstructing the Neckface IR camera data from the Stimulus round using NeckNet-18. The RGB-based datasets (RGBData and OpenData) are obtained through undersampling and sample matching to NeckData.

5.3 Error Detection Model: Results

Table 3 presents the results from the best-performing models for each training method and dataset type. The best-performing model on the NeckData dataset can be seen in Figure 5. *Pre-trained OpenData* (resp. NeckData) refers to

Table 2. Samples (frames) per dataset per type of reaction.

Dataset	Neutral (0)	Error (1)	Total
NeckData	51745	37815	89560
OpenData	51580	37815	89395
NeckFaceIR	51745	37815	89560
RGBData	51578	37815	89393

the model pre-trained on NeckData (resp. OpenData) and fine-tuned on OpenData (resp. NeckData). All time-series models (NeckData and OpenData datasets) performed best in the non-normalized dataset, except for pre-trained and single-participant OpenData (PCA dataset). A full list of hyperparameters tested can be found in the study repository. In sum, our results for error detection from human facial cues are:

- *Time-series models* (trained on OpenFace features or Blendshapes): the model trained on NeckData outperforms the model trained using OpenFace features, by 5% accuracy and 10% F1.
- *Pre-trained Models* (trained on one dataset and fine-tuned on the other): performance is lower than single-dataset training. Model fine-tuned on NeckData outperforms model fine-tuned on OpenData, by 5% accuracy and 5% F1.
- *Single-Participant Models* (trained on a subset of a participant's data and tested on the remaining data from that participant): for the smaller training set (5% of a participant's data), the NeckData model (GRU_FCN) outperforms the model trained using OpenFace features (InceptionTime) by 6% accuracy and F1 (84.7% vs. 78.8% accuracy). For the larger training set (45%), both models exceed 95% accuracy, and the OpenData model slightly outperforms the NeckData model by 1% accuracy and F1 (96.5% vs. 95.5% accuracy); the advantage of NeckData is therefore concentrated in the low-data regime. The NeckData model achieves accuracy above 84% when trained on only 5% of data for a single participant.
- *Frame-Based Models* (trained on images, RGB videos from front-facing camera or videos from the IR cameras on the NeckFace device): model trained on NeckFace camera data outperforms RGB-trained ResNet34 by 1.5% accuracy but has lower F1 (2%).

6 Discussion

Motivated by human social competence, where we continuously interpret external social cues from our peers to navigate social environments, our study explored a novel approach to more contextually aware robots through error detection by leveraging human facial reactions captured through a neck-mounted device, NeckFace. Our contributions are twofold: 1) we adapted NeckNet [8] to a new dataset of human reactions, using a lighter model; 2) we developed models using data from the NeckFace device and NeckNet to detect errors from human reactions, finding good model performance both across and within participant data and outperforming conventional methodologies such as OpenFace.

6.1 3D Facial Mapping Model

The performance for NeckNet-18 was lower than that of the original NeckFace work [8]. This difference may be attributed to the following factors: 1) the original NeckFace work used more calibration data (7-minute rounds) while our calibration session had a duration of 5 minutes, of which only 4 minutes were initially used for training; 2) the original NeckFace device sampled at 30 fps while our device sampled at 12 fps. This is due to our use of a single Raspberry Pi device that records data from two cameras instead of using two Pi's. While implying a lower 3D mapping

Table 3. Model architectures and test set performance for the best-performing model per training method and dataset type. Metrics are shown in $M \pm SD$ for 5 non-overlapping participant folds, or for all participants for single-participant models ($n = 25$). Models using data collected through the RGB camera are shown in grey rows. For single-participant models, results are shown for training-set sizes of 5% and 45% of each participant's data.

Time-series Models					
Model	**Dataset**	**Accuracy**	**Precision**	**Recall**	**F1**
gMLP	**OpenData**	0.606 ± 0.051	0.563 ± 0.025	0.632 ± 0.072	0.535 ± 0.015
GRU_FCN	**NeckData**	0.658 ± 0.061	0.649 ± 0.060	0.667 ± 0.052	0.637 ± 0.070
Pre-trained Models					
gMLP	**OpenData**	0.569 ± 0.040	0.547 ± 0.028	0.572 ± 0.075	0.538 ± 0.019
gMLP	**NeckData**	0.619 ± 0.057	0.597 ± 0.042	0.619 ± 0.057	0.589 ± 0.047
Single-Participant Models – 5% of participant data for training					
InceptionTime	**OpenData**	0.788 ± 0.049	0.782 ± 0.048	0.785 ± 0.050	0.782 ± 0.049
GRU_FCN	**NeckData**	0.847 ± 0.048	0.840 ± 0.048	0.846 ± 0.050	0.842 ± 0.049
Single-Participant Models – 45% of participant data for training					
InceptionTime	**OpenData**	0.965 ± 0.019	0.963 ± 0.020	0.965 ± 0.020	0.964 ± 0.020
GRU_FCN	**NeckData**	0.955 ± 0.028	0.954 ± 0.029	0.954 ± 0.029	0.954 ± 0.029
Frame-Based Models					
ResNet34	**RGB**	0.515 ± 0.025	0.505 ± 0.020	0.505 ± 0.018	0.493 ± 0.023
ResNet34	**NeckFace_IR**	0.530 ± 0.031	0.504 ± 0.027	0.501 ± 0.016	0.474 ± 0.014

performance, these two adjustments were made to minimize the burden on participants regarding the duration of the session and device size, while preserving valid performance. According to Chen et al. [8], when MAE is under 40, the reconstructions should appear visually similar to the ground truth.

6.2 Error Detection Model

We further expanded on prior work on robot error detection through human social cues by extensively exploring the potential of NeckFace-generated data in reaction classification. When comparing cross-participant performance, a GRU model trained on *NeckData* outperforms a gMLP model trained on *OpenData*, i.e., facial action units data extracted from video frames of the participants' reactions recorded on a participant-facing camera, achieving a test accuracy of 65.8%. While low generalizability to unseen participants has been discussed [4], highlighting the need for larger datasets of human reactions to different error contexts, this performance is also higher than reported in previous works [33, 39]. Notably, an experimental factor which could impact this result is different levels of participant engagement. For example, Parreira et al. [33] collected data online through Prolific [2], wherein participants might be distracted or less engaged than when watching videos in-lab. Nonetheless, these results validate NeckFace as a device that effectively captures human reactions and NeckNet as a data reconstruction methodology in the context of this user study.

Interestingly, the pre-trained models (those trained on one dataset— *OpenData, NeckData*—and fine-tuned on the other) did not outperform single-dataset models. Still, we find that a model trained on OpenFace data and fine-tuned using NeckFace data outperforms a model trained on OpenFace data only. This indicates that the NeckFace approach may offer benefits even when utilizing transfer learning techniques. For frame-based models, interestingly, performance was comparable across the two datasets (video frames from an RGB participant-facing camera and from NeckFace's IR camera), with both datasets generating models which perform slightly above chance.

The most striking results come from the single-participant models. Using just 5% of the data for training, the GRU model trained on NeckFace data achieved an accuracy of 84.7%, surpassing the InceptionTime model trained

[2]https://www.prolific.com/

on OpenData (78.8%). This gap closes when using 45% of the data for training (95.5% for NeckData vs. 96.5% for OpenData, Table 3), suggesting that NeckFace is most useful in contexts with few calibration data, allowing for effective personalized error detection systems. Further, the GRU model used only has around 300,000 trainable parameters, which yields potential for lightweight computation and system portability.

The performance of NeckFace-based models, especially in single-participant scenarios, highlights the potential of this approach for creating more adaptable and personalized robot interaction systems. By capturing facial cues from a unique angle, NeckFace appears to provide rich, informative data that can be leveraged for error detection, namely through mapping reactions into a 3D plane rather than OpenFace's 2-dimensional processing. By tapping into previously underutilized data sources, we can enhance robots' ability to interpret human reactions and adjust their behavior accordingly.

It is worth discussing further why models trained on NeckData outperform those trained on OpenData, given that OpenFace is a more standard, widely validated method for extracting facial features from frontal RGB video. We see two plausible, non-exclusive explanations. First, NeckData represents facial expressions as a 3D vector of Blendshape and head-rotation parameters, whereas OpenFace's Action Units are derived from a 2D frontal projection of the face; the additional information captured in the 3D representation may carry signal relevant to error reactions (e.g., head movements toward or away from the scene) that is attenuated or lost in a 2D projection. Second, NeckFace's sensors are positioned beneath the chin, close to the skin, and are therefore less sensitive to the participant's distance from or angle relative to a fixed camera than a frontal RGB camera is; this could make NeckData comparatively more robust to the kind of head movement and pose variation that participants naturally exhibit while reacting to a stimulus. We note, however, that NeckData and OpenData are derived from two different sensing modalities (a custom neck-mounted IR camera versus a standard frontal RGB camera) and processed via two different feature-extraction pipelines (NeckNet-18 versus OpenFace); because more than one variable differs between these two conditions, this comparison cannot isolate which specific factor (sensor modality, feature-extraction method, or the 3D versus 2D representation) drives the performance difference.

We also note that, as a binary classification problem, the cross-participant model performance reported is comparatively modest relative to the much higher accuracy achieved by single-participant models. This pattern is consistent with the broader difficulty of generalizing error detection models to unseen participants given the relatively small and idiosyncratic nature of individual facial reactions [4], and suggests that near-term applications of this approach may be best suited to settings where some degree of per-user calibration data is available (as in our single-participant results) rather than fully participant-agnostic deployment. Future improvements to cross-participant performance could come from larger and more diverse training datasets spanning a wider range of error types, contexts, and individual reaction styles, as well as from feature representations or modeling approaches explicitly designed to be robust to inter-individual variability in reaction style.

6.3 General System Considerations

Detecting and responding to social errors is a critical component of creating socially competent robots that can seamlessly integrate into diverse human environments. Our use of human facial reactions as a signal for error detection builds directly from HRI literature [4, 34, 42]. While the technical approach leverages machine learning, the core objectives and implications of our work are deeply rooted in human-robot interaction applications. The inclusion of human and robot error videos in the stimulus dataset explores the potential of a generalized error detection system which would allow agents to contextualize failures in their environment. Future work may explore higher error granularity,

e.g., through detecting robot error type [43].To the best of our knowledge, no prior work has explored human reactions for *multi-class* error detection of this kind in the environment, which could improve the robot's social and functional performance; we note that related work has studied multi-dimensional human annotation of robot performance along axes such as task proficiency and social appropriateness [47], or error events in successive robot error detection [28].

We acknowledge that the proposed system implies additional steps and computation, including the calibration round and the 3D reconstruction using NeckNet-18 before error detection. This work aimed to push the boundaries of input data modalities used in HRI systems by exploring the NeckFace device and its functionalities. We verify that error detection accuracy in models trained on NeckData is higher by 12-15% than when directly using frame-based datasets. Similar results on frame-based models for non-overlapping participant testing have been reported [4, 33]. While these are non-exhaustive results and there is potential for RGB-based models to detect error effectively, higher computational demands for feature extraction might be justified by better-performing systems. This tradeoff warrants discussion, especially in more critical application scenarios.

Taken together, we see this work as offering two preliminary, but useful, takeaways for researchers working on wearable or alternative sensing modalities for HRI. First, a 3D, neck-mounted facial reaction signal can match or exceed the performance of an established 2D, frontal-camera-based pipeline (OpenFace) for binary error detection in this dataset, which suggests that reaction-sensing systems need not be tied to a frontal-camera form factor to be useful. Second, the gap between cross-participant and single-participant model performance observed here echoes a broader pattern in this area of HRI research: with limited training data, generic, participant-agnostic error detection remains hard, while lightweight, quickly-personalized models can perform comparatively well [4, 33]. We offer these as exploratory observations from a single, relatively small-scale study rather than prescriptive design guidelines, given the scope and limitations discussed in Section 6.4.

6.4 Limitations and Future Work

It is important to highlight the scope and limitations of our work. The controlled environment of our user study may not fully represent the complexities of real-world human-robot interactions. For example, NeckFace is highly sensitive to lighting variations, which can compromise the performance of NeckNet. Further data collection in different environmental conditions, as well as additional research (e.g., in-person errors) is needed to validate the effectiveness of this approach across diverse populations and in various real-world scenarios, where noisy reactions are common. The present study evaluates NeckFace in a stationary, video-elicitation paradigm rather than during an in-person, mobile interaction. While this design choice allowed us to benchmark NeckFace's facial reaction data against established alternatives under controlled, matched conditions, it means that our evaluation does not yet directly test the mobility-related motivation for using a wearable, neck-mounted sensor in the first place. We see this as a necessary first step rather than a complete validation of NeckFace's intended use case, and a natural next step for future work would be to evaluate NeckFace during an embodied, in-person interaction in which participants are co-located and moving with the robot.

Beyond the controlled setting itself, our experimental design has methodological limitations that should be considered when interpreting our comparison across data sources. Model selection was not performed identically across all model families: most models were selected based on average test accuracy across all 5 cross-validation folds, but ResNet34-based frame models (NeckFaceIR and RGBData) were selected based on test accuracy on a single fold, due to the substantially higher computational cost of training ResNet34 on full video frames across all folds. This is a potential confound when comparing ResNet34-based results to the other model families, and revisiting ResNet34 model selection with the same

5-fold protocol used elsewhere is an item for future work. The cross-participant estimates in Table 3 rest on a small number of individuals per fold and should be read as indicative rather than precise; the comparatively large standard deviations across folds reflect this. Finally, Table 3 reports only the best-performing architecture per protocol and dataset. The margin between that architecture and the alternatives we searched was not systematically characterized, so our findings should not be read as evidence that a particular architecture is best suited to this task; they concern the data source, which was compared under a common search procedure for both NeckData and OpenData.

We also reflect on what our findings imply for using NeckFace in practice, given several practical issues we encountered during data collection. First, the device requires an explicit calibration step for each new wearer, which adds a few minutes of setup time before each use; this may be acceptable for some deployment contexts (e.g., a robot's primary user calibrating the device once per session or per day) but is a real overhead to account for in others. Second, facial accessories such as large earrings could shake during wear and disturb image quality, leading us to exclude 2 participants' data; future hardware iterations may need to account for, or be robust to, common facial and head accessories. Third, several participants experienced inadequate NeckFace camera adjustment during data collection, leading to a further 3 exclusions; this points to a need for either more robust, self-adjusting camera mounts, or simple in-session feedback mechanisms that let a wearer or researcher know when the device is misaligned, particularly in less-supervised, real-world deployment settings where a researcher is not present to manually adjust the device as we did in this lab study. The exclusion of participants from our dataset due to the inability to effectively transform the NeckFace camera data onto the 3D reconstruction brings about the tradeoff between larger datasets versus better-curated datasets. A longer calibration round or higher data frame rate could also provide more accurate NeckFace data reconstruction.

Finally, we discuss the ethical and privacy implications of this work specifically. Because NeckFace is intended to capture and interpret an observer's or user's facial reactions, its use raises privacy and consent considerations distinct from many other HRI sensing approaches: it records detailed facial expression data about a person who may not always be the primary, consenting subject of a given interaction (e.g., a bystander who did not opt in to being monitored). Any deployment of this technology should be accompanied by clear consent procedures and limits on how reaction data is stored, used, and for how long, particularly if reactions to a robot's behavior could be linked back to an identifiable individual. While the envisioned use-case – more harmonious human-robot interactions, where robots play a collaborative and complementary role to human activities and needs,— should contribute to increased human satisfaction, fulfillment, and efficiency, there is a risk of misuse of this technology. Broader discussions ought to be held to ensure frameworks where new tools are not misappropriated and contribute to a more just and equitable society. Related to this, access to technology is also an important indicator of development and equity. The NeckFace device is relatively inexpensive and easy to replicate, and while its current version is not yet optimized for low power consumption, newer and more efficient materials should mitigate this limitation.

7 Conclusion

This research underscores our claim that expanding human-in-the-loop robot sensing can lead to more adaptable and personalized robot interaction systems. Leveraging the social competence humans exhibit through interpreting social cues, NeckFace offers a new perspective on capturing these previously untapped data sources. The performance of NeckFace-derived models, especially in single-participant scenarios where personalization is key, reinforces its potential for creating more context-aware robots. By effectively capturing facial cues, NeckFace could eventually enable robots to better understand and respond to human reactions in real time. This capability is essential for integration into dynamic human environments, pushing robots beyond simple task execution toward becoming socially aware collaborators

guided by human behavior. The NeckFace form factor and classification results give confidence for further exploration into other HRI domains, where traditional methods have previously been limited.

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